Organic Process Research & Development

Rapid Enabling of Negishi Couplings for a Pair of mGluR5 Negative Allosteric Modulators

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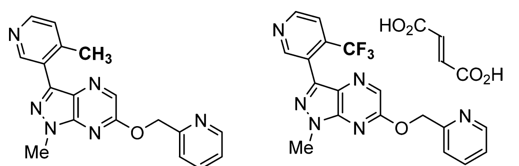
Supporting Information

ABSTRACT: A pair of end game Negishi reactions was developed for the large-scale synthesis of two mGluR5 negative allosteric modulators from a common template. Key developmental aspects include the preparation of the core iodide and organozinc reagents, the management of an exotherm during cross-coupling by switching from a batch process to a semibatch process, and the removal of metals (Pd, Zn, Mg) from the Negishi products. Various aqueous workups, carbon treatments, and Si-Thiol were screened for the purging of heavy metals to regulatory limits, a task made more challenging by the metal coordinating abilities of both drug substances. In the case of one API, a series of crystallizations was required to provide the drug substance with the desired purity and solid form. Both of these Negishi couplings were implemented on hectogram to kilogram scale under the aggressive timelines of early process development.

■ INTRODUCTION

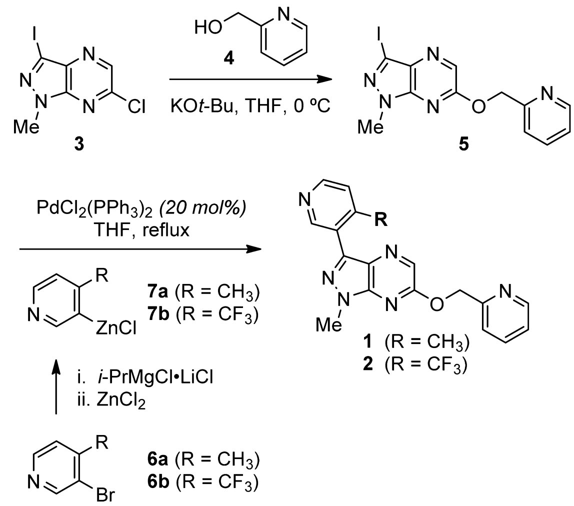
Metabotropic glutamate receptor 5 negative allosteric modulators (mGluR5 NAMs) are promising therapies for various conditions, including depression, anxiety, Fragile X syndrome, addiction, and L-DOPA-induced dyskinesia.[1] Recently our medicinal chemistry group reported on the discovery of several mGluR5 NAM clinical candidates, many of which are analogs prepared in divergent fashion from a common core.[2] A pair of these compounds, **1** and **2**, matriculated into process development once it was decided that hectogram to kilogram quantities of each would be required for preclinical and clinical studies. Both **1** and **2** share a common pyrazolopyrazine core and 2-pyridylmethoxy tail, and differ structurally only with respect to the methyl versus trifluoromethyl substituent on the picoline headgroup. Compound **2** was developed as the fumarate salt to avoid any complications from milling and formulating the free base, a relatively low melting solid (mp 92 °C).



1 **2•fumarate**

Our medicinal chemistry group prepared both active pharmaceutical ingredients (APIs) from pyrazolopyrazine **3**,[2] a core template which could be elaborated modularly at the chloride via S_NAr displacement and at the iodide via transition-metal-catalyzed cross-coupling (Scheme 1). The initial procedure for chloride displacement treated **3** with 2-pyridinemethanol (**4**) and KO*t*-Bu in THF. The resulting ether was purified via aqueous workup and the trituration of crude solids in EtOAc. Subsequent Negishi coupling of iodide **5** with pyridylzinc chloride **7a** or **7b** (2 equiv) completed the synthesis of mGluR5 NAMs **1** and **2**. The requisite zinc reagents were prepared by treating the corresponding bromides **6a–b** with *i*-PrMgCl·LiCl[3] and $ZnCl_2$, and their couplings with the iodide proceeded with a high catalyst loading of 20 mol% $PdCl_2(PPh_3)_2$ in THF at reflux. After aqueous workups, multiple chromatographic separations were required to purify both **1** and **2**. Alternatively, our medicinal chemistry group also explored Suzuki couplings of **5** with the corresponding pinacol boronic esters; however, these employed higher temperatures and pressures (dioxane at 160 °C) and proved less efficient overall.[2]

Scheme 1. Medicinal Chemistry Synthesis of 1 and 2



While the medicinal chemistry synthesis allowed rapid access to several analogs from a common core, the long linear sequence to pyrazolopyrazine **3** (six steps from 5-amino-1-methylpyrazole)[2] diminished the appeal of this overall route for

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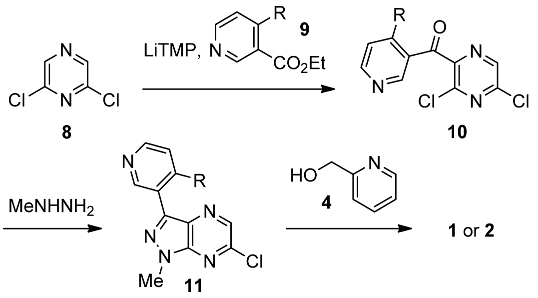
clinical manufacturing. Aggressive delivery timelines only gave us a very brief opportunity for route exploration, and we considered more convergent approaches to **1** and **2** that could provide kilograms of API. Scheme 2 shows one promising

Scheme 2. More Convergent Route to 1 and 2 via Pyrazine Acylation and Pyrazole Annulation[2]

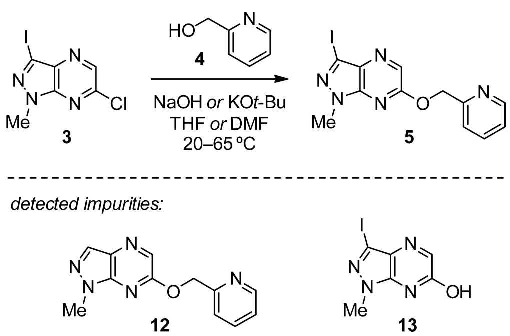


strategy involving the acylation of 2,6-dichloropyrazine (**8**)[4] with picolinate **9** followed by pyrazole annulation and chloride displacement with 2-pyridinemethanol (**4**).[2] Ultimately, we needed to order raw materials before this new route was fit for this purpose, and we defaulted to sourcing pyrazolopyrazine **3** as GMP starting material and enabling the medicinal chemistry route to **1** and **2** for kilogram scale. Specifically, this entailed developing scalable processes for the preparation of iodide **5** and its Negishi couplings with pyridylzinc reagents **7a** and **7b**. The cross-couplings, in particular, would require substantial development to ensure robustness at lower catalyst loadings and to deliver high purity API without resorting to chromatography. Furthermore, various aqueous workups and metal scavengers would be evaluated for the purging of metals (Pd, Zn, Mg) to regulatory limits, which we anticipated might be a challenge due to the coordinating abilities of both **1** and **2**.

■ PREPARING IODIDE 5

Iodide **5** was prepared from pyrazolopyrazine **3** via base-mediated chloride displacement by 2-pyridinemethanol (Scheme 3). Medicinal chemists used KO*t*-Bu and THF as

Scheme 3. Exploring Conditions for Chloride Displacement



the base and solvent for this etherification; however, our cursory exploration of these conditions provided the S_NAr product with up to 7% of desiodo byproduct **12**.[5] A screen of alternate bases identified 50% aqueous NaOH, which did not generate impurity **12** and was preferred as an inexpensive and moisture insensitive alternative to KO*t*-Bu. The etherification in DMF with 2 equiv of 50% NaOH at 45 °C proceeded quickly (>99% conversion after 10 min), but the product **5** was unstable under the reaction conditions and hydrolyzed to **13** (16% after 25 min). Reaction in THF under phase-transfer conditions resulted in slower conversion (50% after 10 min) but negligible decomposition of **5** to **13** (<1% after 15 h), and thus, the etherification in THF was developed further for kilogram scale.

An added benefit of THF was the opportunity to crystallize iodide **5** directly from this solvent upon aqueous dilution. To aid in the design of this direct-drop isolation,[6] the solubility of etherification product in various THF/H_2O mixtures was measured. Whereas saturated solutions in THF contained 41 mg of **5** per gram of solution, the solubility fell to less than 10 mg/g when diluting with an equal volume of H_2O and to near-zero solubility in 2:1 w/w H_2O/THF (Figure 1). In gram-scale

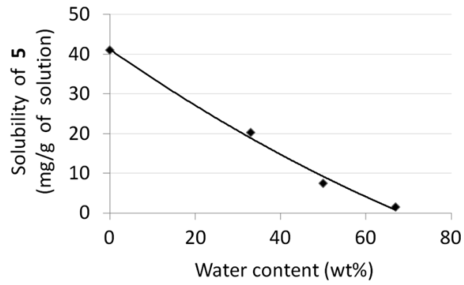
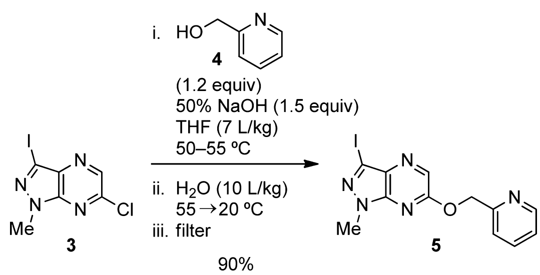


Figure 1. Solubility of iodide **5** in THF/H_2O at 20 °C.

pilots, aqueous dilution of reaction mixtures to 60:40 w/w H_2O/THF provided **5** with the proper balance of yield (>90%) and purity (>99%), and this workup/isolation strategy was incorporated into our scaleup.

The optimized process for chloride displacement to iodide **5** on kilogram scale is shown in Scheme 4. A solution of chloride **3** and alcohol **4** (1.2 equiv) in THF was charged with 1.5 equiv of 50% aq NaOH solution and heated until <1% unreacted **3**. Whereas laboratory batches held at 60 °C proceeded to >99% conversion after several hours, we were required to operate at a slightly lower temperature (50−55 °C) on kilogram scale to comply with our internal safety regulations for overnight processing. As a result, the reaction had only progressed to ∼90% conversion after 10 h, and supplemental charges of **4** (0.1 equiv and 0.05 equiv) were added to expedite consumption of remaining chloride. The product **5** was crystallized by diluting the mixture with H_2O and cooling to 20 °C. The solids were collected in a filter, and residual NaOH was purged by rinsing the cake with water until the eluting filtrate was pH-neutral. From this direct-drop isolation, 2.63 kg of **5** were isolated in 90% yield and 99.5% purity.

Scheme 4. Optimized Process for Chloride Displacement to Iodide 5

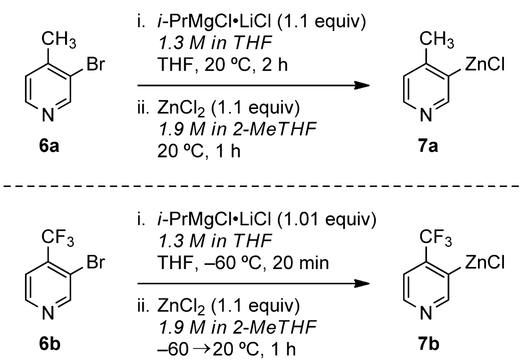


■ NEGISHI COUPLINGS

With iodide **5** in hand, we turned our attention to its reaction with picoline coupling reagents to complete the synthesis of mGluR5 NAMs **1** and **2**. As previously noted, our medicinal chemistry colleagues found that Negishi couplings[7] of **5** with pyridylzinc reagents **7a** and **7b** were superior to Suzuki couplings of **5** with the corresponding pinacol boronic esters. Traditionally, Negishi couplings have been less common than Suzuki couplings on large scale,[8] as organozinc reagents require some operational complexity to prepare, demonstrate some functional group incompatibilities, and generate substantial zinc waste. However, with limited time to deliver bulk quantities of **1** and **2**, we decided to enable the pair of Negishi couplings in parallel and focus on developing workups and purifications that would provide high purity API without resorting to chromatography. We anticipated that heavy metal (Pd, Zn) contamination of **1** and **2** might be a concern due to the various Lewis basic sites within both drug substances, and the development of effective metal purging strategies from these Negishi products would be a high priority.

i. Formation of Pyridylzinc Reagents 7a and 7b. Pyridylzinc reagents **7a** and **7b** were prepared from the corresponding bromides **6a** and **6b** via metal−halogen exchange with *i*-PrMgCl·LiCl[3] and transmetalation to $ZnCl_2$. Whereas the medicinal chemistry group employed 1.3 equiv of magnesium and zinc reagents relative to the bromide in THF at −78 °C, we found that only a very slight excess (1.01−1.1 equiv) of either reagent was sufficient and resulted in less metal waste (Scheme 5). In the case of **6a**, metal−halogen exchange

Scheme 5. Metalation of Bromides to Pyridylzinc Reagents



with only 1.1 equiv of *i*-PrMgCl·LiCl at 20 °C proceeded to completion within 2 h (as monitored by HPLC for the conversion of **6a** to debrominated picoline). In the case of **6b**, metal−halogen exchange was much more rapid and proceeded at −60 °C with only 1.01 equiv of *i*-PrMgCl·LiCl. A mild exotherm accompanied the addition of this magnesium reagent to either bromide (~5 °C increase over 5 min addition). Treating the pyridylmagnesium intermediates with $ZnCl_2$ provided **7a** and **7b**, and this transmetalation produced a stronger exotherm that was mediated by slow addition and jacketed cooling. Zinc chloride was dosed as a commercial solution to avoid handling hygroscopic solids, and in our experience, $ZnCl_2$ solutions in 2-MeTHF proved more reliable than $ZnCl_2$ solutions in THF.[9] Our prepared solutions of **7a** and **7b** were homogeneous and were carried directly into the Negishi coupling.

For the preparation of **1** on kilogram scale, we identified a commercial source of **7a**: Rieke Metals, Inc.[10] manufactures bulk quantities of 4-methyl-3-pyridylzinc bromide[11] (0.5 M in THF) by metalation of bromopicoline with activated zinc. In Negishi coupling, this commercial reagent performed similarly to the organozinc reagent prepared by transmetalation, and its application in our process obviated the need to purge Mg salts from the reaction mixture.

ii. Coupling of Pyridylzinc Reagents and Iodide 5. The typical Negishi protocol from medicinal chemistry mixed the prepared solution of **7a** or **7b** with iodide **5** and $PdCl_2(PPh_3)_2$ in THF at room temperature, and then heated the combined mixture to reflux. This combination of catalyst and solvent performed well on small scale and was retained for further development. After an iterative screen of the reaction conditions, we were able to reduce the amount of pyridylzinc reagent from 2.0 equiv to 1.4−1.5 equiv and the catalyst loading from 20 mol% to 2−5 mol% without any sacrifice in conversion.

While developing the Negishi coupling, we discovered that reaction of **7a** (1.5 equiv) and **5** with the catalyst produced a substantial exotherm which occurred when the reaction mixture was being heated from room temperature to reaction temperature and thus remained unnoticed during early smaller-scale experiments. As depicted in Figure 2, the internal

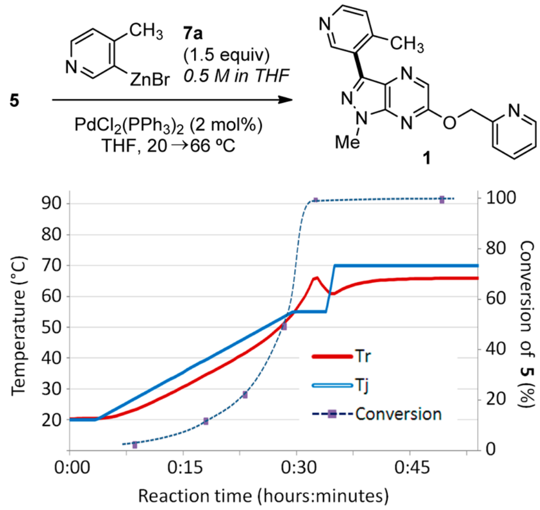


Figure 2. Original batch process for Negishi coupling of **7a** and **5**.

reactor temperature (T_r) increased from 50 °C to reflux (~66 °C) over the span of 5 min despite maintaining a jacket temperature (T_j) of 55 °C. This coincided with a surge in reaction conversion, and once the exotherm subsided, the Negishi coupling had essentially proceeded to completion. The risk of an uncontrolled, runaway exotherm in the batch mode prompted us to develop a semibatch process where the reaction rate would be controlled by the addition rate of one of the reagents. In a revised process (Figure 3), the solution of **5** and catalyst was heated at 55−60 °C and **7a** was dosed gradually while maintaining the batch temperature in the desired range to ensure fast reaction rate and avoid accumulation of unreacted reagents. There was a noticeable delay of a few minutes between the start of pyridylzinc addition and the beginning of the exotherm development. Under these conditions, the Negishi coupling had proceeded to near completion by the time the pyridylzinc reagent was fully charged and there was no change in the product purity profile compared to the batch process. This semibatch process was implemented twice on

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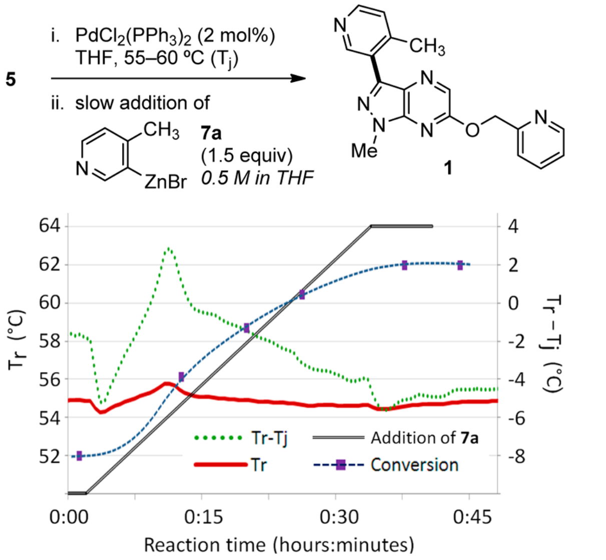
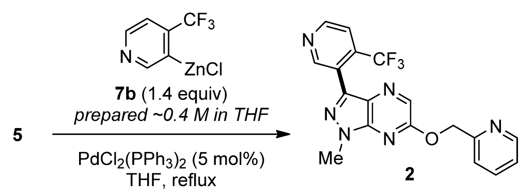


Figure 3. Revised semibatch process for Negishi coupling with slow addition of **7a**.

kilogram scale in which the reagent was charged continuously over 30 min; the heat evolution was mild and the batch temperature peaked at 3 °C above the jacket temperature during the course of addition.

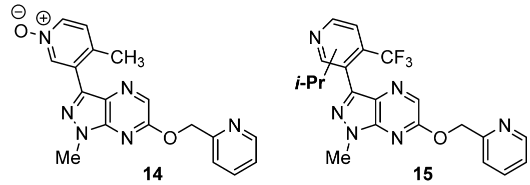
No exotherm was measured in the parallel Negishi reaction of fluorinated analog **7b** and iodide **5**, and the original protocol of premixing the coupling partners prior to heating was retained for a couple of batches above 100-g scale (Scheme 6). The

Scheme 6. Negishi Coupling of 7b and 5



pyridylzinc reagent (1.4 equiv) was prepared from the bromide according to the *i*-PrMgCl·LiCl/$ZnCl_2$ protocol outlined in Scheme 5 and telescoped directly into the coupling. Whereas pilots on 20-g scale proceeded to completion within 4 h at reflux, reactions were noticeably slower above 100-g scale and stalled after 19 h with 8–9% remaining **5**. These stalled reactions were driven to completion with second charges of **7b** (0.3 equiv) and catalyst (2 mol%). Interestingly, this incomplete conversion on 100-g scale was observed using the same lots of reagents and solvent and the same precautions to exclude oxygen that performed well on 20-g scale. Factors contributing to this discrepancy are under investigation.[12]

The Negishi couplings for the synthesis of **1** and **2** both proceeded fairly cleanly. In addition to the expected side products from the quenching and homocoupling of the pyridylzinc reagents, reaction mixtures contained only low levels of other impurities including pyridine oxide **14** (<1%), isopropyl adduct **15** (2%), triphenylphosphine oxide (2%), and, in the case of **1**, an impurity with a mass consistent with brominated API (1–2%).[13] The pyridine oxide was not detected as a side product in **2**, which led us to speculate that oxidation occurred in **1** at the more electron-rich picoline. The isopropyl regiochemistry of adduct **15** was not determined, but we suspect that isopropyl radicals derived from *i*-PrZnCl (from 1% excess *i*-PrMgCl relative to **6b**) or the byproduct *i*-PrBr reacted at the electron-deficient picoline.[14]



14 **15**

iii. Metal Purging and Purification of 1 and 2.

As mentioned previously, one drawback of Negishi couplings is the generation of stoichiometric zinc waste. Furthermore, when magnesium reagents such as *i*-PrMgCl are used to prepare the organozinc reagent, the byproduct Mg salts often complicate aqueous workups due to limited solubility under basic pH.[15] In the cases of **1** and **2**, the coordinating abilities of both APIs presented an added challenge to metal purging. As the residual levels of metals in drug substances must be controlled to regulatory limits,[16] we needed to develop a workup and purification that would separate Pd, Zn, and Mg from the Negishi products.

The metal-scavenging efficiencies of several aqueous workups were evaluated for the Negishi reaction mixtures. An initial strategy involved exploiting the basicity of **1** and **2** by extracting the API into aqueous HCl. We believed this approach would have two benefits: (1) nonbasic impurities (e.g., trace **5**, PPh_3) would be retained in the organic layer, and (2) the protonation of Lewis basic sites in **1** and **2** would interfere with API coordination to metals. However, neutralization of the HCl extraction layer with NaOH precipitated the API with 2% Pd and 10–20% Zn (and Mg, in the case of **2**). Alternatively, washing the crude API solution with aqueous EDTA[17] (mixture of disodium and trisodium salts) purged Pd to 1360 ppm and Zn and Mg to <30 ppm. Aqueous NH_4OH proved more effective at scavenging Pd (to 630 ppm) and equally effective at scavenging Zn and Mg (to <30 ppm). As such, crude Negishi mixtures of **1** were charged directly with NH_4OH while cooling at 10 °C to mitigate an exotherm from quenching.

A similar quench of **2** with aqueous NH_4OH resulted in the precipitation of Mg salts throughout the aqueous phase. To avoid the laborious filtration of these salts, an alternative workup was developed to keep them in aqueous solution. First, the crude Negishi mixture was quenched with aqueous NH_4Cl, which provided two homogeneous layers for clean phase separation and had the added benefit of scavenging some Zn and Mg. Compound **2** was then extracted from the organic layer into aqueous HCl, and then this acidic layer was neutralized with NH_4OH in the presence of THF/MTBE. The result of this pH swing[15] was two homogeneous phases in which **2** partitioned back to the organic layer and residual Zn and Mg were retained in the aqueous layer.

The aqueous workups sufficiently purged Zn and Mg from the Negishi products, but residual Pd was still a problem. In the case of **2**, we were encouraged that crystallization (after aqueous workup) from EtOAc/heptane reduced Pd by 10-fold; however, even with this reduction, we suspected that crystallization alone would not sufficiently reject this heavy metal to regulatory limits. Thus, several activated carbons and Si-Thiol were screened for their ability to scavenge Pd from **2**, with the goal of identifying conditions that would be effective

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for **1** as well (Table 1). As a general trend, all of the surveyed treatments were more effective Pd scavengers at 50 °C than

Table 1. Activated Carbons and Si-Thiol to Scavenge Metals from 2

treatment[a]	Pd	Zn	Mg
none	847[b]	19	18
Darco KBB, 20 °C	517	2	7
Darco KBB, 50 °C	238	8	14
Darco G60, 20 °C	739	4	8
Darco G60, 50 °C	510	8	8
Norit E Supra, 20 °C	682	6	10
Norit E Supra, 50 °C	524	11	13
Si-Thiol, 20 °C	87	2	4
Si-Thiol, 50 °C	**26**	**6**	**18**

[a]Added 25 wt% scavenger to a solution of **2** in EtOAc (10 mL/g). [b]Values in ppm for concentrated sample.

20 °C, and Si-Thiol emerged as the superior treatment for Pd removal. With these results, the thiol resin was incorporated into the workups of **1** and **2** after the aforementioned NH_4OH extractions of Zn (and Mg). Hence, washed organic solutions of crude API were treated with Si-Thiol at 50 °C, and two batches of **1** (in THF) on kilogram scale were purified with 25 wt% resin to 4 and 11 ppm Pd. A similar treatment on 230 g of **2** (in EtOAc after solvent exchange) with 43 wt% resin provided API with >400 ppm Pd. The higher residual metal in **2** stemmed in part from the higher catalyst loading, but we were not concerned, as subsequent fumarate salt formation and crystallization would provide another opportunity for Pd purging (vide infra).

After aqueous workup and Si-Thiol treatment, the remaining challenge was isolating the drug substances in the correct solid form while purging the lingering organic impurities that were not washed away during aqueous workup. Both solutions of **1** and **2** at this point contained <2% of any single impurity including unreacted iodide **5**, triphenylphosphine oxide, pyridine oxide **14**, isopropyl adduct **15**, and brominated **1**. The procedures for the final isolation became very compound specific and will be described separately.

Compound **1** was formulated as the unsolvated free base (Form 1, most thermodynamically stable form identified to date). The desired form could be obtained by cooling crystallization with proper seeding from a variety of solvents, of which isopropyl acetate (*i*-PrOAc) was chosen based on the solubility data (Figure 4) and its classification as a Class 3 solvent in the ICH guidelines.[18] Unfortunately, the crystallization of Form 1 resulted in very poor impurity rejection, which would necessitate multiple recrystallizations to achieve the desired purity level. On the other hand, a dihydrate of **1** could be easily isolated from THF after dilution with water (Figure 5), and that isolation did provide the desired impurity

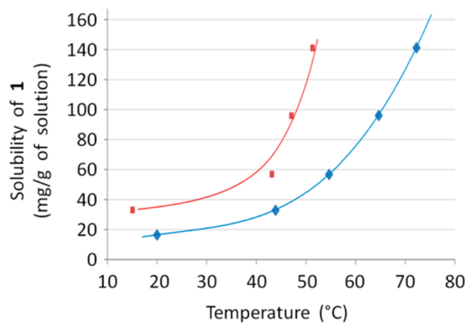


Figure 4. Solubility of **1** (Form 1) in *i*-PrOAc as a function of temperature (blue line) and metastable zone of the supersaturated solution (red line).

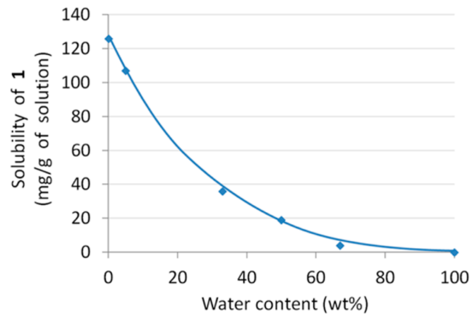
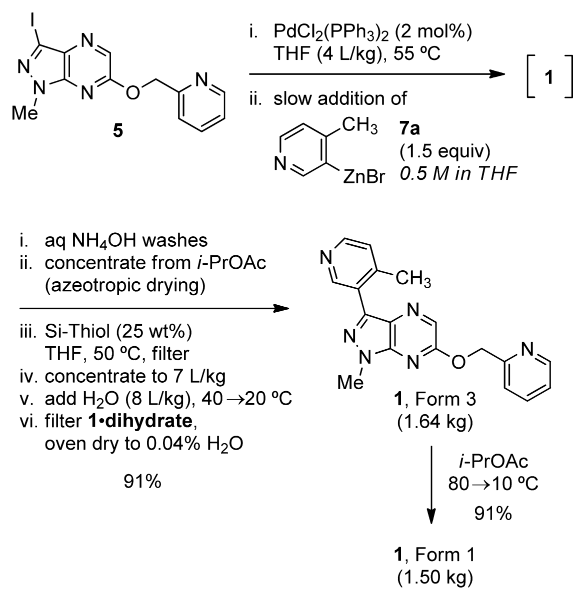


Figure 5. Solubility of **1·dihydrate** in THF/H_2O at 20 °C.

purge. The isolated solid dehydrated easily upon drying in a vacuum oven, resulting in another anhydrous form (Form 3), distinct from the desired Form 1. The required conversion of Form 3 to Form 1 was carried out by heating the isolated Form 3 in *i*-PrOAc until dissolved, cooling the solution slightly to obtain a metastable supersaturated solution (metastable zone between the curves in Figure 4), seeding the batch with Form 1, and cooling the resulting slurry slowly to 20 °C. Interestingly, an attempt was made to skip the seeding of the batch (to avoid the additional effort of generating GMP seed material), which resulted in haphazard crystallization of the initial Form 3. The correct Form 1 was obtained by reheating the solution to complete dissolution and conducting the recrystallization with seeding. The execution of the recrystallization in the kilo lab facility afforded 1.50 kg of **1** as Form 1 (99.8% purity, 83% overall yield, Scheme 7).

API **2** was being formulated as a fumarate salt which was prepared after initial crystallization of the free base. After treating the crude solution of **2** with Si-Thiol, organic

Scheme 7. Overall Negishi Process for the Synthesis of 1



impurities were purged by crystallizing the free base from EtOAc and heptane. The resulting solids were a mixture of what appeared to be two different crystal habits: finer, pale solids (~5%) and denser, darker solids (95%) (Figure 6). The two crystal types contained drastically different amounts of residual Pd: the pale solids contained 1910 ppm while darker ones only 408 ppm. Repeated heating and cooling cycles failed to produce a homogeneous crystal bed; instead the mixture of filtered solids was taken into the salt formation step. Crystallization of the fumarate salt from EtOAc/MeOH resulted in the desired purge of residual Pd to 40 ppm and afforded 230 g of **2·fumarate** as a uniform solid (99.9% purity, 71% overall yield, Scheme 8).

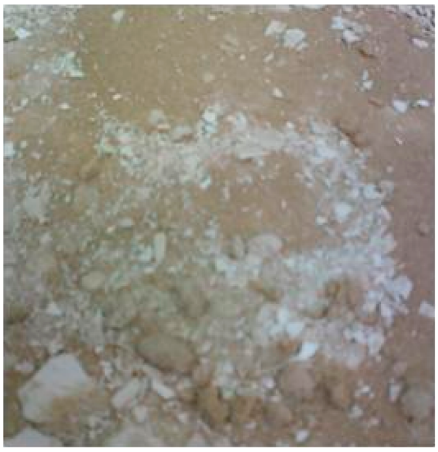
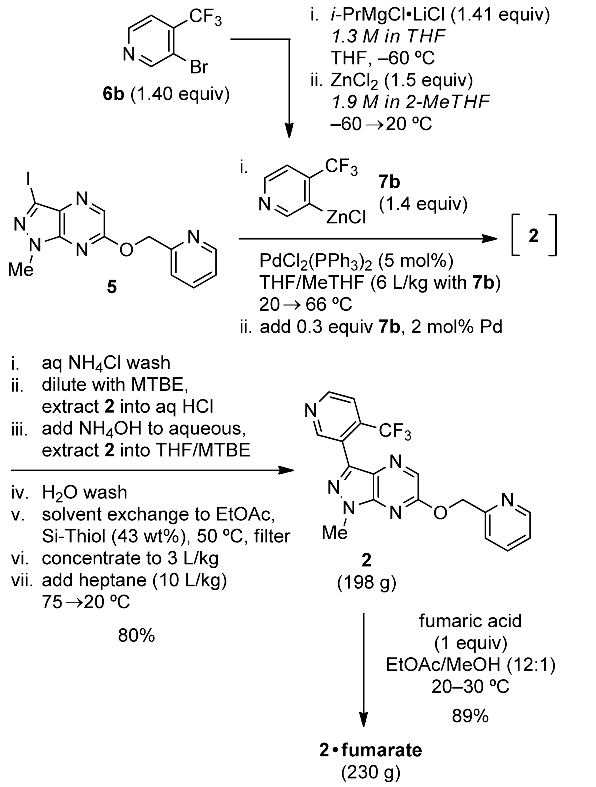


Figure 6. Heterogeneous crystals of free base **2** prior to fumarate salt formation.

Scheme 8. Overall Negishi Process for the Synthesis of 2·fumarate



CONCLUSIONS

In this manuscript, we described the development of a pair of end game Negishi reactions for the synthesis of mGluR5 NAMs **1** and **2** (Schemes 7 and 8). Each step was discussed in detail, including the preparation of iodide **5** and pyridylzinc reagents **7a** and **7b**, the coupling of these reagents via Pd catalysis, the evaluation of aqueous workups and scavengers for metal purging, and the handling of solid form issues for the crystallization of high purity API. These Negishi reactions were successfully implemented on kilogram and hectogram scale.

EXPERIMENTAL SECTION

The in-process HPLC analyses were carried out on an Agilent 1100 or a similar liquid chromatography system capable of binary solvent delivery and equipped with an FID detector. A C18 reverse phase column (Atlantis T3 3-μm 4.6 mm × 150 mm or equivalent) was utilized with an acetonitrile−water gradient (modified with 0.05% TFA) from 0 to 100% MeCN. The signal was collected and reported at 210 or 215 nm. The retention times of the reaction components were determined by injecting the authentic markers. The purity of the isolated intermediates and final API was determined on a Waters Acquity H-Class UPLC system using a BEH C8 1.7 μm, 2.1 mm × 100 mm column and a linear gradient of 5 to 95% MeOH with a 10 mM aq $NaHCO_3$ solution over 8.5 min. The signal was recorded at 210 nm.

3-Iodo-1-methyl-6-(pyridin-2-ylmethoxy)-1*H*-pyrazolo[3,4-*b*]pyrazine (5). A stirred solution of chloride **3** (2.35 kg, 7.98 mol, 1.00 equiv) and 2-pyridinemethanol (**4**) (1.05 kg, 9.62 mol, 1.20 equiv) in THF (16 L) at 20 °C was charged with 50% NaOH solution (0.96 kg, 12 mol, 1.5 equiv) and heated at 50−55 °C for 10 h (11% unreacted **3**). A second charge of 2-pyridinemethanol (0.090 kg, 0.82 mol, 0.10 equiv) was added, and the mixture was stirred at 50−55 °C for 12 h (1% unreacted **3**). A third and final charge of 2-pyridinemethanol (0.045 kg, 0.41 mol, 0.050 equiv) was added, and the mixture was stirred at 50−55 °C for 5 h. The mixture was diluted with H_2O (23 L) and cooled to 20 °C over 2 h. The solids were collected on a Nutsche filter, rinsed with H_2O until the eluting filtrate was pH neutral (28 L), and dried at 45 °C under vacuum with a N_2 sweep to obtain 2.63 kg (90% yield) of **5** as a white solid: UPLC 99.5%; mp 155−156 °C; ^1H NMR (400 MHz, d_6-DMSO): 8.59 (d, J = 4.4 Hz, 1H), 8.34 (s, 1H), 7.85 (app t, J = 7.6 Hz, 1H), 7.60 (d, J = 7.6 Hz, 1H), 7.37 (app t, J = 6.0 Hz, 1H), 5.57 (s, 2H), 3.95 (s, 3H); ^{13}C NMR (100 MHz, d_6-DMSO): 158.8, 155.3, 149.2, 140.9, 137.0, 133.9, 130.1, 123.2, 122.2, 94.0, 68.9, 34.0; HRMS-ESI m/z: $[M + H]^+$ calcd for $C_{12}H_{10}IN_5O$, 368.0003; found, 368.0002.

1-Methyl-3-(4-methylpyridin-3-yl)-6-(pyridin-2-ylmethoxy)-1*H*-pyrazolo[3,4-*b*]pyrazine (1). A stirred solution of iodide **5** (1.00 kg, 2.72 mol, 1.0 equiv) and $PdCl_2(PPh_3)_2$ (38 g, 0.054 mol, 0.020 equiv) in THF (4 L) was heated at 55 °C and charged with a solution of 4-methyl-3-pyridylzinc bromide (**7a**) (0.50 M in THF, 8.0 kg, 4.1 mol, 1.5 equiv) over 30 min. The resulting mixture was heated at 55 °C for 10 min and then cooled to 10 °C over 2 h. Saturated aqueous NH_4OH (2.7 L) was added over 20 min while maintaining the internal temperature below 25 °C. The biphasic mixture was further diluted with H_2O (1.3 L) and agitated at 15 °C for 10 min. The phases were separated, the aqueous layer was back-extracted with THF (1.5 L), and the combined organic layers were washed sequentially with saturated aqueous NH_4OH (1.7 L) and 20% aqueous NaCl (2 L). The organic layer was concentrated to 3 L volume and then dried azeotropically by diluting with *i*-PrOAc (9 L) and reconcentrating to 2 L volume twice. The mixture was diluted with THF (10 L) to dissolve all the solids, stirred over Si-Thiol (250 g) at 20 °C for 8 h, and passed through a pad of Celite in a Nutsche filter with a THF rinse (4 L). A sample of filtrate was concentrated to crude solids of **1** and measured for heavy metals: 11 ppm Pd, 49 ppm Zn.

The above procedure was repeated on the same scale to provide a second batch of crude **1** filtrate. A sample from this second batch was concentrated to crude solids and measured for heavy metals: 4 ppm Pd, 38 ppm Zn. Both batches were combined, concentrated to 13 L volume, and held at 40 °C as H_2O (15 L) was added over 30 min. The resulting slurry was cooled to 20 °C over 2 h and granulated at 20 °C for another 2 h. The solids were collected on a Nutsche filter, rinsed with 2:1 v/v H_2O/THF (8 L), and dried at 50 °C under vacuum with a N_2 sweep to obtain 1.64 kg (91% yield) of **1** (Form 3) as a white solid (UPLC 99.6%).

The entire batch of solids was dissolved in *i*-PrOAc (8 L) at 75 °C and then held at 60 °C until seeds formed. The thin slurry was held at 65 °C for 30 min and cooled to 10 °C over 3 h. PXRD of the crystals revealed Form 3. A 25 mL sample of the Form 3 slurry was recrystallized in a separate flask by dissolving the solids at 80 °C, cooling to 70 °C, and holding for 30 min as crystals formed, and then cooling to 20 °C over 4 h. This segregated slurry contained Form 1 seeds which would be added to the parent batch. Thus, the parent slurry of Form 3 in *i*-PrOAc was reheated at 75 °C to dissolve the solids, cooled to 67 °C, and charged with the Form 1 seeds. Once the seed bed was established, the slurry was cooled to 10 °C over 3 h and granulated at 10 °C for 10 h. PXRD of the crystals confirmed desired Form 1. The solids were collected on a Nutsche filter, rinsed with 2:1 v/v H_2O/THF (8 L), and dried at 50 °C under vacuum with N_2 sweep to obtain 1.50 kg (91% yield for recrystallization; 83% yield from **5**) of **1** (Form 1) as a white solid: UPLC 99.8%; mp 114−115 °C; ^1H NMR (400 MHz, d_6-DMSO): 9.01 (s, 1H), 8.60 (d, J = 4.4 Hz, 1H), 8.47 (d, J = 4.8 Hz, 1H), 8.41 (s, 1H), 7.85 (dt, J = 1.6, 7.6 Hz, 1H), 7.61 (d, J = 8.0 Hz, 1H), 7.37 (d, J = 4.8 Hz, 1H), 7.36 (d, J = 5.2 Hz, 1H), 5.59 (s, 2H), 4.02 (s, 3H), 2.52 (s, 3H); ^{13}C NMR (100 MHz, d_6-DMSO): 158.0, 155.5, 150.0, 149.2, 148.9, 145.4, 141.2, 139.6, 137.0, 133.5, 127.0, 126.1, 125.8, 123.2, 122.2, 68.7, 33.9, 20.4; HRMS-ESI m/z: $[M + H]^+$ calcd for $C_{18}H_{16}N_6O$, 333.1458; found, 333.1461.

1-Methyl-6-(pyridin-2-ylmethoxy)-3-(4-(trifluoromethyl)pyridin-3-yl)-1H-pyrazolo[3,4-*b*]pyrazine fumarate (2·fumarate). A stirred solution of bromopyridine **6b** (107 g, 472 mmol, 1.40 equiv) in THF (470 mL) was cooled at −70 °C as a solution of *i*-PrMgCl·LiCl (1.30 M in THF, 365 mL, 475 mmol, 1.41 equiv) was added via addition funnel over 20 min while maintaining the internal temperature below −60 °C. The resulting brown mixture was stirred at −70 °C for 10 min. A solution of $ZnCl_2$ (2.06 M in 2-MeTHF, 245 mL, 505 mmol, 1.50 equiv) was added via addition funnel over 20 min while maintaining the internal temperature below −60 °C, and the resulting mixture was allowed to warm to 20 °C over 1 h. The resulting pyridylzinc reagent **7b** was treated with iodide **5** (124 g, 337 mmol, 1.00 equiv), $PdCl_2(PPh_3)_2$ (11.5 g, 16.4 mmol, 0.048 equiv), and THF (120 mL), and the mixture was heated at reflux (70 °C internal) for 19 h (8% unreacted **5**). Second charges of $PdCl_2(PPh_3)_2$ (4.4 g, 6.3 mmol, 0.019 equiv) and **7b** (0.30 equiv) were added, and the reaction mixture was heated at reflux for 2 h and cooled to 20 °C. Saturated aqueous NH_4Cl (1.6 L) was added, and the biphasic mixture was stirred for 30 min. The organic layer was washed with saturated aqueous NH_4Cl (1.6 L), diluted with MTBE (1.3 L), and extracted with 1 M aqueous HCl (1.3 L). The aqueous phase was washed with 1:1 v/v THF/MTBE (1.3 L) and then stirred over another portion of 1:1 v/v THF/MTBE (1.3 L), as saturated aqueous NH_4OH (1.3 L) was added dropwise via addition funnel over 20 min. The resulting biphasic mixture was stirred at 20 °C for 15 h. The organic layer was separated and diluted with EtOAc (1.3 L), washed twice with H_2O (1.3 L per wash), and concentrated to a dark oil of crude **2**.

A second batch of crude **2** was prepared in parallel following the above procedure (including second charges of catalyst and **7b**) scaled to 112 g of iodide **5**, and the two batches were combined and reconcentrated twice from EtOAc to 230 g of dark brown solids. The crude solids were dissolved in EtOAc (2.4 L) and stirred over Si-Thiol (100 g) at 50 °C for 5 h and then 20 °C for 16 h. The slurry was filtered through Celite, the filter pad was rinsed with EtOAc (1.0 L), and the filtrate was concentrated to 750 mL volume. Heptane (1.0 L) was added, and the solids were dissolved upon heating to 75 °C. The solution was cooled to 40 °C at a rate of 1 °C/min (seeds formed), and additional heptane (1.0 L) was added at a rate of 30 mL/min. The resulting slurry was cooled to 20 °C at a rate of 1 °C/min, and additional heptane (0.5 L) was added at a rate of 30 mL/min. The mixture was stirred at 20 °C for 6 h and filtered through a Buchner funnel. The filter cake was rinsed with heptane (1.0 L) and dried at 50 °C under vacuum with a N_2 sweep to obtain 198 g (80% yield) of **2** as a mixture of pale and medium brown solids.

The entire batch of **2** (198 g, 513 mmol) was dissolved in EtOAc (2.9 L), charged with a solution of fumaric acid (59.5 g, 513 mmol) in MeOH (1.5 L), and stirred at 20 °C for 1 h. The mixture containing fine solids was concentrated to 2 L volume under vacuum (50−70 Torr, 20−30 °C bath), and then distillation was continued at constant volume with EtOAc addition (~2 L) until the mother liquor was a 92:8 molar ratio of EtOAc/MeOH (^1H NMR). The resulting slurry was stirred at 20 °C for 20 h and filtered through a Buchner funnel. The filter cake was rinsed with heptane (1.0 L) and dried at 50 °C under vacuum with a N_2 sweep to obtain 230 g (89% yield from **2**) of **2·fumarate** as a pale beige solid: UPLC >99.9%; ICP-MS: 40 ppm Pd, 1946 ppb Mg, 801 ppb Zn; mp 154−155 °C; ^1H NMR (400 MHz, d_6-DMSO): 13.12 (br s, 2H), 9.09 (s, 1H), 8.95 (d, J = 5.2 Hz, 1H), 8.59 (d, J = 4.4 Hz, 1H), 8.42 (s, 1H), 7.94 (d, J = 5.2 Hz, 1H), 7.85 (app t, J = 8.0 Hz, 1H), 7.62 (d, J = 7.6 Hz, 1H), 7.37 (app t, J = 6.0 Hz, 1H), 6.63 (s, 2H), 5.61 (s, 2H), 4.04 (s, 3H); 1.007/1.000 molar ratio of fumaric acid to **2**; ^{13}C NMR (100 MHz, d_6-DMSO): 166.0, 158.3, 155.4, 152.8, 151.1, 149.3, 141.1, 137.4, 137.0, 135.0 (q, J_{CF} = 32 Hz), 134.4, 134.0, 126.4, 124.2 (q, J_{CF} = 2 Hz), 123.2, 122.7 (q, J_{CF} = 273 Hz), 122.2, 120.5 (q, J_{CF} = 5 Hz), 68.8, 34.1; HRMS-ESI m/z: $[M + H]^+$ calcd for $C_{18}H_{13}F_3N_6O$, 387.1176; found, 387.1170.

■ ASSOCIATED CONTENT

Supporting Information

[1]H and [13]C NMR spectra for **5**, **1**, and **2·fumarate**, and PXRD patterns for **1** (Forms 1 and 3). This material is available free of charge via the Internet at http://pubs.acs.org.

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Notes

The authors declare no competing financial interest.

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